



中國海洋大學海德學院

HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

Course Introduction

Statistical Modelling and Inference

Instructor: Dr. Waqar Hafeez

Spring 2026



中國海洋大學海德學院

HAIDE COLLEGE, OCEAN UNIVERSITY OF CHINA

About your Instructor

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Major: Statistics

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Lectures

Two lectures per week:

Tuesday 8:00am - 9:50am

(week 1-5, 9-13)

Friday 3:30pm - 5:20pm

(week 1-14)

All lecture slides will be posted on BB.

Theoretical Lecture

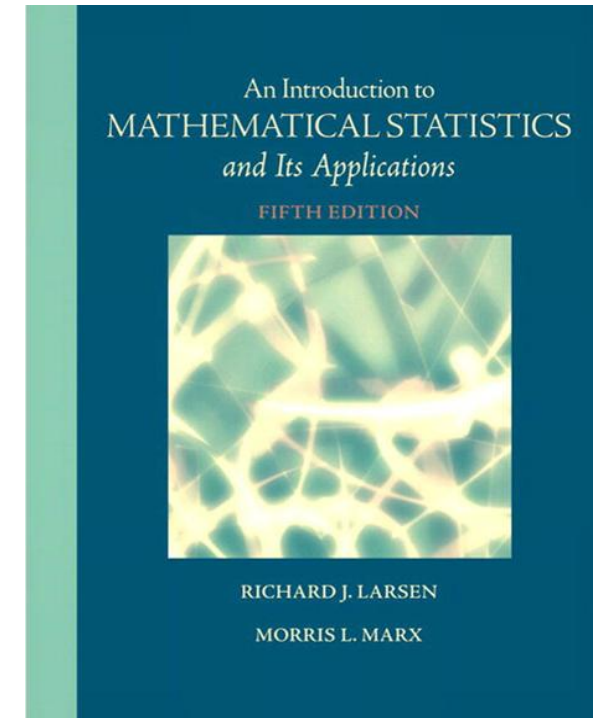
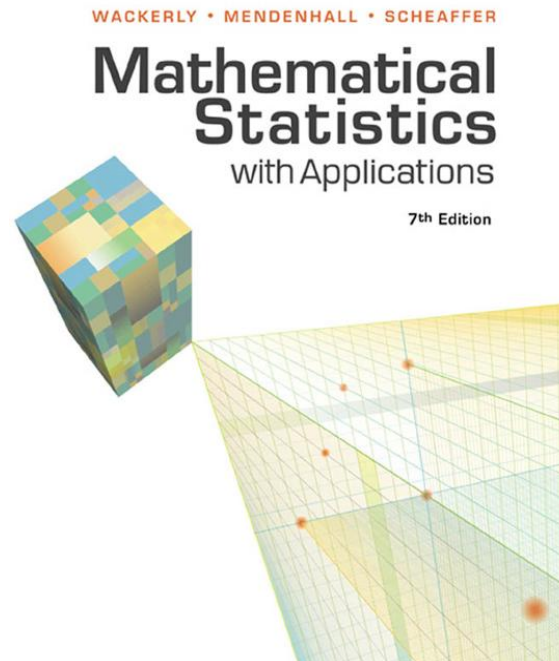
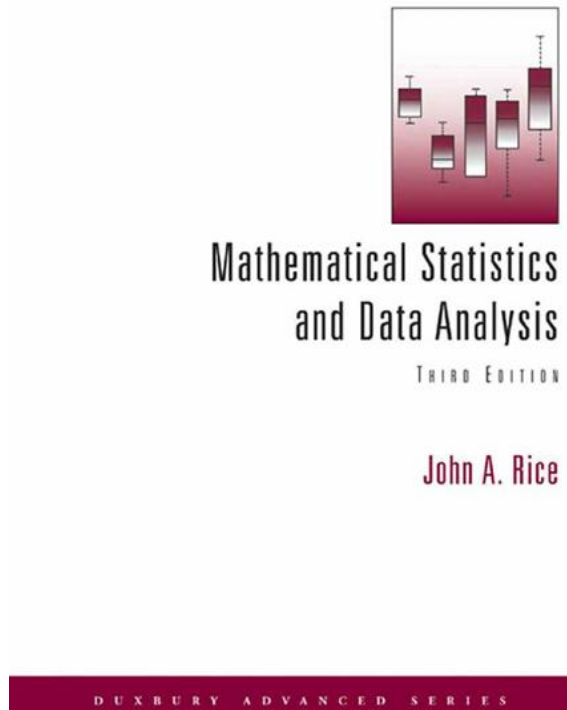
Practical Lecture

Course Outline

1. Introduction to statistical inference: notation, mean squared error
2. Best Linear Unbiased Estimation (BLUE)
3. Confidence intervals, tests of hypotheses and power calculations
4. Inference for a single sample, unknown variance; pivotal quantities
5. Inference for two independent samples
6. Regression modelling and least squares estimation
7. Prediction for regression and residuals
8. Multiple linear regression and least squares estimation
9. BLUE and tests of hypotheses
10. Applications to prediction, polynomial regression and one-way analysis of variance
11. Analysis of covariance and two-way analysis of variance
12. Maximum likelihood (ML) estimation
13. Inference for ML estimators and tests based on the likelihood
14. Practical issues: randomization, imputation, multiple testing, non-parameteric tests

Reference textbooks

1. J.A. Rice: Mathematical Statistics and Data Analysis, 3rd edition (2007).
2. D.D. Wackerly, W. Mendelhall and R.L. Scheaffer: Mathematical Statistics with Applications, 7th edition (2008).
3. R.J. Larsen, M.L. Marx: An Introduction to Mathematical Statistics and its Applications, 5th edition (2012).



Assessment Summary

Assessment task	Weighting	Due
Quizzes	10%	Weeks 1-7, 9-11
Assignments	20%	Weeks 3,5,7,11
Mid-semester Test - In-person, Invigilated	15%	Week 8, during the scheduled Seminar
Examination	55%	Examination period

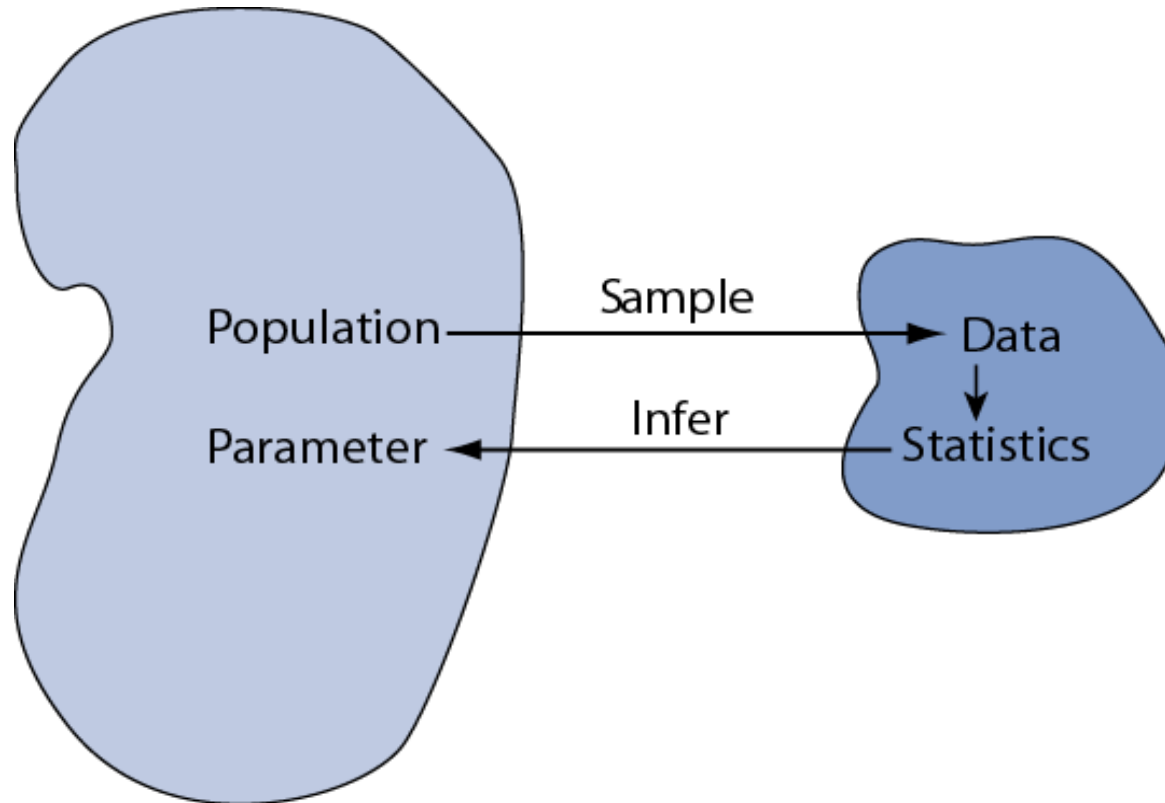
Lecture 1, Outline

- Introduction to statistical inference
- Notation
- Point estimates
- Unbiasedness
- Mean squared error

Introduction to statistical inference

Statistical inference is the act of generalizing from a **sample** to a **population** with calculated degree of certainty.

We want to
learn about
*population
parameters*
...



using
statistics
calculated
in the
sample

Population

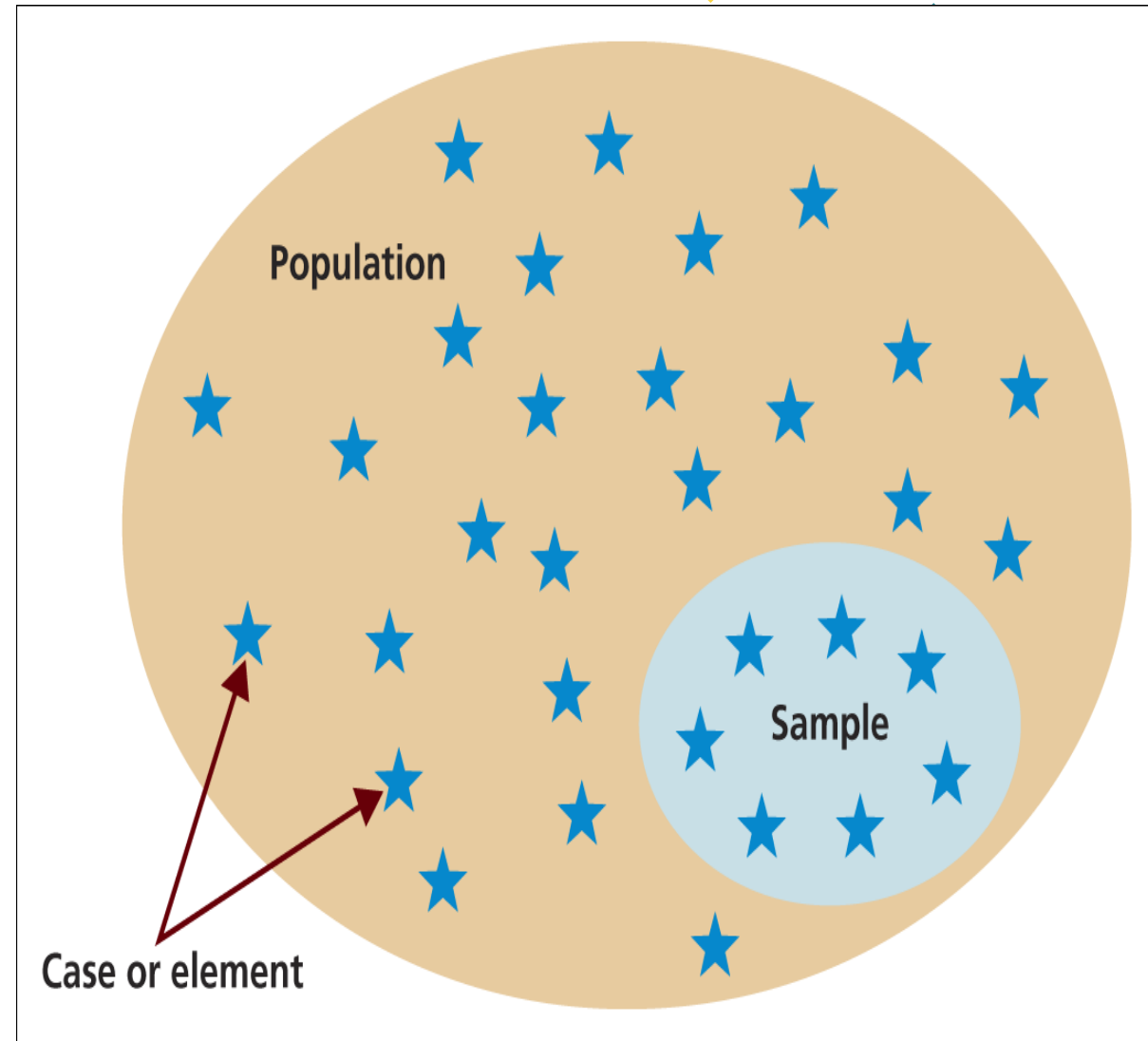
A population is a collection of objects or units about which we want to know something or draw an inference. The population may be finite or infinite. The population is denoted by S or P and its size by N.

Note: Estimates from population are called population parameters. e.g., Mean (μ), Variance (σ^2) and Standard deviation (σ).

Sample

A subset of the population, which represents the entire population, is called a sample. The sample is denoted by s and its size is denoted by n.

Note: Estimates from sample are called sample statistic. e.g., Mean ($\bar{x}$), Variance (s^2) and Standard deviation (s).



Parameters and Statistics

We MUST draw a distinction between
parameters and statistics

	Parameters	Statistics
Source	Population	Sample
Calculated?	No	Yes
Constants?	Yes	No
Examples	μ, σ, ρ	$\bar{x}, s, \hat{p}$

Point estimates

A point estimate is a single value calculated from sample data that serves as the "best guess" for an unknown population parameter.

Estimator

An estimator is a rule, often expressed as a formula, that tells how to calculate the value of an estimate based on the measurements contained in a sample.

i. Mean

The *mean* of a sample of n measured responses $y_1, y_2, \dots, y_n$ is given by

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i.$$

The corresponding population mean is denoted μ .

ii. Variance

The *variance* of a sample of measurements $y_1, y_2, \dots, y_n$ is the sum of the square of the differences between the measurements and their mean, divided by $n - 1$. Symbolically, the sample variance is

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2.$$

The corresponding population variance is denoted by the symbol σ^2 .

iii. Standard deviation

The *standard deviation* of a sample of measurements is the positive square root of the variance; that is,

$$s = \sqrt{s^2}.$$

The corresponding *population* standard deviation is denoted by $\sigma = \sqrt{\sigma^2}$.

The Bias and Mean Square Error of Point Estimators

i. Bias

Let $\hat{\theta}$ be a point estimator for a parameter θ .

if $E(\hat{\theta}) = \theta$, then $\hat{\theta}$ is an unbiased estimator of θ .

if $E(\hat{\theta}) \neq \theta$, then $\hat{\theta}$ is said to be biased estimator of θ .

The **bias** of a point estimator $\hat{\theta}$ is given by $B(\hat{\theta}) = E(\hat{\theta}) - \theta$.

ii. Mean Square Error

The mean square error of a point estimator $\hat{\theta}$ is

$$MSE(\hat{\theta}) = E [(\hat{\theta} - \theta)^2]$$

The mean square error of an estimator $\hat{\theta}$, is a function of both its variance and its bias. If $B(\hat{\theta})$ denotes the bias of the estimator $\hat{\theta}$, MSE can be shown that

$$MSE(\hat{\theta}) = V(\hat{\theta}) + [B(\hat{\theta})]^2$$

Properties Expectation and Variance

i. $E(c) = c$

ii. $E(c\hat{\theta}) = cE(\hat{\theta})$

iii. $E(\hat{\theta} \pm c) = E(\hat{\theta}) \pm c$

iv. $E[E(\hat{\theta})] = E(\hat{\theta})$

v. $Var(\hat{\theta}) = E(\hat{\theta} - E(\hat{\theta}))^2 = E(\hat{\theta}^2) - [E(\hat{\theta})]^2$

vi. $Var(c) = 0$

vii. $Var(a\hat{\theta}) = a^2Var(\hat{\theta})$

viii. $Var(\hat{\theta} \pm c) = Var(\hat{\theta})$

ix. $Var(a\hat{\theta} \pm c) = a^2Var(\hat{\theta})$

Exercise 1

Using this identity

$$(\hat{\theta} - \theta) = [\hat{\theta} - E(\hat{\theta})] + [E(\hat{\theta}) - \theta] = (\hat{\theta} - E(\hat{\theta})) + B(\hat{\theta})$$

Show That

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = V(\hat{\theta}) + [B(\hat{\theta})]^2.$$

Proof:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

$$MSE(\hat{\theta}) = E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2] = E\left[\left((\hat{\theta} - E(\hat{\theta})) + B(\hat{\theta})\right)^2\right]$$

Because $B(\hat{\theta}) = E(\hat{\theta}) - \theta$

$$MSE(\hat{\theta}) = E\left[(\hat{\theta} - E(\hat{\theta}))^2 + (B(\hat{\theta}))^2 + 2\left((\hat{\theta} - E(\hat{\theta}))B(\hat{\theta})\right)\right]$$

$$MSE(\hat{\theta}) = E\left[(\hat{\theta} - E(\hat{\theta}))^2\right] + E[B(\hat{\theta})^2] + 2 \cdot E[\hat{\theta} - E(\hat{\theta})] \cdot B(\hat{\theta})$$

Exercise 1

$$MSE(\hat{\theta}) = E \left[\left(\hat{\theta} - E(\hat{\theta}) \right)^2 \right] + E \left[B(\hat{\theta})^2 \right] + 2 \cdot E \left[\hat{\theta} - E(\hat{\theta}) \right] \cdot B(\hat{\theta})$$

$$MSE(\hat{\theta}) = Var(\hat{\theta}) + [B(\hat{\theta})]^2$$

Because: $E[\hat{\theta} - E(\hat{\theta})] = E(\hat{\theta}) - E(\hat{\theta}) = 0$

and $V(\hat{\theta}) = E \left(\hat{\theta} - E(\hat{\theta}) \right)^2$

Exercise 2

a. If $\hat{\theta}$ is an unbiased estimator of θ , what is $B(\hat{\theta})$?

Solution:

Given that $E(\hat{\theta}) = \theta$

Bias is

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta$$
$$B(\hat{\theta}) = 0$$

b. If $B(\hat{\theta}) = 5$, what is $E(\hat{\theta})$?

Solution:

Therefore,

$$B(\hat{\theta}) = E(\hat{\theta}) - \theta$$
$$\Rightarrow 5 = E(\hat{\theta}) - \theta$$
$$\Rightarrow E(\hat{\theta}) = 5 + \theta$$

Exercise 3

Suppose that $\hat{\theta}$ is an estimator for a parameter θ and $E(\hat{\theta}) = a\theta + b$ for some nonzero constants a and b .

a. In term of a , b and , what is $B(\hat{\theta})$?

Solution:

Because

$$\begin{aligned} B(\hat{\theta}) &= E(\hat{\theta}) - \theta \\ B(\hat{\theta}) &= a\theta + b - \theta \end{aligned}$$

Hence

$$B(\hat{\theta}) = (a - 1)\theta + b$$

Exercise 3

Suppose that $\hat{\theta}$ is an estimator for a parameter θ and $E(\hat{\theta}) = a\theta + b$ for some nonzero constants a and b .

b. Find a function of $\hat{\theta}$, let say $\hat{\theta}^*$ that is an unbiased estimator of θ .

Solution:

Because $\hat{\theta}^*$ that is an unbiased estimator of θ ,

$$\begin{aligned}\Rightarrow B(\hat{\theta}^*) &= 0 \\ \Rightarrow E(\hat{\theta}^*) - \theta &= 0 \\ \Rightarrow E(\hat{\theta}^*) &= \theta\end{aligned}$$

Given that $E(\hat{\theta}) = a\theta + b$

Gives us, $\theta = \frac{E(\hat{\theta}) - b}{a}$

$$\begin{aligned}\Rightarrow E(\hat{\theta}^*) &= \frac{E(\hat{\theta}) - b}{a} \\ \Rightarrow \hat{\theta}^* &= \frac{\hat{\theta} - b}{a}\end{aligned}$$

Because a and b are constants.

$\Rightarrow E(a) = a$ and $E(b) = b$

Exercise 4

Refer to **Exercise 1**.

a. If $\hat{\theta}$ is an unbiased estimator of θ , how does $MSE(\hat{\theta})$ compare to $V(\hat{\theta})$?

Solution: As we know that

$$MSE(\hat{\theta}) = V(\hat{\theta}) + [B(\hat{\theta})]^2$$

because $\hat{\theta}$ is an unbiased estimator of $\theta \Rightarrow B(\hat{\theta}) = 0$

$$\begin{aligned}\Rightarrow MSE(\hat{\theta}) &= V(\hat{\theta}) + [0]^2 \\ \Rightarrow MSE(\hat{\theta}) &= V(\hat{\theta})\end{aligned}$$

b. If $\hat{\theta}$ is biased estimator of θ , how does $MSE(\hat{\theta})$ compare to $V(\hat{\theta})$?

Solution:

because $\hat{\theta}$ is biased estimator of $\theta \Rightarrow B(\hat{\theta}) \neq 0$

$$\Rightarrow MSE(\hat{\theta}) = V(\hat{\theta}) + [B(\hat{\theta})]^2$$

Exercise 5

Suppose that $E(\hat{\theta}_1) = E(\hat{\theta}_2) = \theta$, $V(\hat{\theta}_1) = \sigma_1^2$ and $V(\hat{\theta}_2) = \sigma_2^2$.

Consider the estimator $\hat{\theta}_3 = a\hat{\theta}_1 + (1 - a)\hat{\theta}_2$.

a. Show that $\hat{\theta}_3$ is an unbiased estimator of θ .

Solution: As given,

$$\begin{aligned}\hat{\theta}_3 &= a\hat{\theta}_1 + (1 - a)\hat{\theta}_2 \\ E(\hat{\theta}_3) &= E(a\hat{\theta}_1 + (1 - a)\hat{\theta}_2) \\ \Rightarrow E(\hat{\theta}_3) &= aE(\hat{\theta}_1) + (1 - a)E(\hat{\theta}_2) \\ &\Rightarrow E(\hat{\theta}_3) = a\theta + (1 - a)\theta \\ &\Rightarrow E(\hat{\theta}_3) = a\theta + \theta - a\theta \\ &\Rightarrow E(\hat{\theta}_3) = \theta\end{aligned}$$

proved that $\hat{\theta}_3$ is an unbiased estimator of θ .

b. If $\hat{\theta}_1$ and $\hat{\theta}_2$ are independent, how should the constant a be chosen to minimize the variance of $\hat{\theta}_3$?

Solution: As given,

$$\begin{aligned}\hat{\theta}_3 &= a\hat{\theta}_1 + (1 - a)\hat{\theta}_2 \\ V(\hat{\theta}_3) &= V(a\hat{\theta}_1 + (1 - a)\hat{\theta}_2)\end{aligned}$$

Exercise 5

$$V(\hat{\theta}_3) = V(a\hat{\theta}_1 + (1-a)\hat{\theta}_2)$$

$$V(\hat{\theta}_3) = a^2V(\hat{\theta}_1) + (1-a)^2V(\hat{\theta}_2)$$

Given that $V(\hat{\theta}_1) = \sigma_1^2$ and $V(\hat{\theta}_2) = \sigma_2^2$

$$V(\hat{\theta}_3) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2$$

$$\Rightarrow V(\hat{\theta}_3) = a^2\sigma_1^2 + (1 + a^2 - 2a)\sigma_2^2$$

$$\Rightarrow V(\hat{\theta}_3) = a^2\sigma_1^2 + \sigma_2^2 + a^2\sigma_2^2 - 2a\sigma_2^2$$

the constant a be chosen to minimize the variance of $\hat{\theta}_3$, take the first derivative and replace equal to zero.

$$\frac{d}{da}V(\hat{\theta}_3) = 0$$

$$\frac{d}{da}(a^2\sigma_1^2 + \sigma_2^2 + a^2\sigma_2^2 - 2a\sigma_2^2) = 0$$

$$\Rightarrow 2a\sigma_1^2 + (0)\sigma_2^2 + 2a\sigma_2^2 - 2\sigma_2^2 = 0$$

$$\Rightarrow a\sigma_1^2 + a\sigma_2^2 - \sigma_2^2 = 0$$

$$\Rightarrow a = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}$$

the required value of constant a , to minimize the variance of $\hat{\theta}_3$ if $\hat{\theta}_1$ and $\hat{\theta}_2$ are independent.

Exercise 5

c. How should the constant a be chosen to minimize the variance of $\hat{\theta}_3$ if $\hat{\theta}_1$ and $\hat{\theta}_2$ are not independent but are such that $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) = c \neq 0$

Solution: As given,

$$\hat{\theta}_3 = a\hat{\theta}_1 + (1 - a)\hat{\theta}_2$$

$$V(\hat{\theta}_3) = V(a\hat{\theta}_1 + (1 - a)\hat{\theta}_2)$$

$$V(\hat{\theta}_3) = a^2V(\hat{\theta}_1) + (1 - a)^2V(\hat{\theta}_2) + 2a(1 - a)\text{Cov}(\hat{\theta}_1, \hat{\theta}_2)$$

Given that $V(\hat{\theta}_1) = \sigma_1^2$, $V(\hat{\theta}_2) = \sigma_2^2$ and $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) = c$

$$V(\hat{\theta}_3) = a^2\sigma_1^2 + (1 - a)^2\sigma_2^2 + 2a(1 - a)c$$

$$\Rightarrow V(\hat{\theta}_3) = a^2\sigma_1^2 + (1 + a^2 - 2a)\sigma_2^2 + 2ac - 2a^2c$$

$$\Rightarrow V(\hat{\theta}_3) = a^2\sigma_1^2 + \sigma_2^2 + a^2\sigma_2^2 - 2a\sigma_2^2 + 2ac - 2a^2c$$

the constant a be chosen to minimize the variance of $\hat{\theta}_3$, take the first derivative and replace equal to zero.

$$\frac{d}{da}V(\hat{\theta}_3) = 0$$

Exercise 5

$$\frac{d}{da}(a^2\sigma_1^2 + \sigma_2^2 + a^2\sigma_2^2 - 2a\sigma_2^2 + 2ac - 2a^2c) = 0$$

$$\Rightarrow 2a\sigma_1^2 + (0)\sigma_2^2 + 2a\sigma_2^2 - 2\sigma_2^2 + 2c - 4ac = 0$$

$$\Rightarrow a\sigma_1^2 + a\sigma_2^2 - \sigma_2^2 + c - 2ac = 0$$

$$\Rightarrow a(\sigma_1^2 + \sigma_2^2 - 2c) = \sigma_2^2 - c$$

$$\Rightarrow a = \frac{\sigma_2^2 - c}{\sigma_1^2 + \sigma_2^2 - 2c}$$

the required value of constant a , to minimize the variance of $\hat{\theta}_3$ if $\hat{\theta}_1$ and $\hat{\theta}_2$ are not independent.

**Thank
You**

谢谢

